

5G RAN

Device-Pipe Synergy Feature Parameter Description

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1

Change History

1.1 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Optimized the equivalent QCI mapping mechanism for service identification-based assurance. For details, see: 10.1 Principles 10.3.2.3 Function Impacts 10.4.1.1 Data Preparation 10.4.1.2 Using MML Commands	Added the gNodeBAlgo.QciAlgoOptSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions of the principles, prerequisite functions, data preparation, and MML-based configurations for service identification-based assurance. For details, see [10.1 Principles](#), [10.3.2.1 Prerequisite Functions](#), [10.4.1.1 Data Preparation](#), and [10.4.1.2 Using MML Commands](#).

1.2 5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 03 (2025-11-07).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

None

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Device-pipe identification	- In NSA networking, device-pipe identification is controlled by the eNodeB, which implements device-pipe negotiation with UEs. - In SA networking, device-pipe identification is controlled by the gNodeB, which implements device-pipe negotiation with UEs.	4 Device-Pipe Identification
Compensation for SAR decrease (low-frequency TDD)	Supported only in SA networking	5 Compensation for SAR Decrease (Low-Frequency TDD)
Fast RRC connection release	Supported only in SA networking	6 Fast RRC Connection Release
BWP spatial adaptation (low-frequency TDD)	Supported only in SA networking	7 BWP Spatial Adaptation (Low-Frequency TDD)
Uplink SINR compensation for non-coherent rank-1 2T UEs	Supported only in SA networking	8 Uplink SINR Compensation for Non-Coherent Rank-1 2T UEs
Coordination between UL skipping and BWP switching	Supported only in SA networking	9 Coordination Between UL Skipping and BWP Switching
Joint service experience assurance	Supported only in SA networking	10 Joint Service Experience Assurance

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high

frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Device-pipe identification	None	4 Device-Pipe Identification
Compensation for SAR decrease (low-frequency TDD)	Supported only in low frequency bands	5 Compensation for SAR Decrease (Low-Frequency TDD)
Fast RRC connection release	Supported only in low frequency bands	6 Fast RRC Connection Release
BWP spatial adaptation (low-frequency TDD)	Supported only in low frequency bands	7 BWP Spatial Adaptation (Low-Frequency TDD)
Uplink SINR compensation for non-coherent rank-1 2T UEs	None	8 Uplink SINR Compensation for Non-Coherent Rank-1 2T UEs
Coordination between UL skipping and BWP switching	Supported only in low frequency bands	9 Coordination Between UL Skipping and BWP Switching
Joint service experience assurance	Supported only in low frequency bands	10 Joint Service Experience Assurance

3 Overview

Device-pipe synergy refers to the synergy between the eNodeB/gNodeB and capable UEs and provides enhanced functions on the eNodeB/gNodeB and UE sides to improve user experience and network performance. [Table 3-1](#) lists the functions involved in device-pipe synergy, whereby device-pipe identification is the basic function and the rest are enhanced functions. All enhanced functions depend on device-pipe identification. Unless otherwise specified, UEs in this document refer to UEs capable of device-pipe synergy.

Table 3-1 Device-pipe synergy functions

Type	Function Name	Introduction	Chapter/Section
Basic function	Device-pipe identification	The eNodeB/gNodeB and UEs exchange signaling to verify each other's support for device-pipe synergy functions.	4 Device-Pipe Identification
Enhanced function	Compensation for SAR decrease (low-frequency TDD)	The gNodeB compensates for channel quality deterioration caused by SAR decrease of UEs.	5 Compensation for SAR Decrease (Low-Frequency TDD)
Enhanced function	Fast RRC connection release	UEs instruct the gNodeB to release RRC connections directly instead of waiting until the inactivity timer expires, thereby reducing UE power consumption.	6 Fast RRC Connection Release

Type	Function Name	Introduction	Chapter/Section
Enhanced function	BWP spatial adaptation (low-frequency TDD)	A UE reports the maximum number of downlink MIMO layers to be used to the gNodeB. After switching to an appropriate power saving BWP, the UE shuts down some receive antennas. The gNodeB performs downlink scheduling based on the UE-reported maximum number of downlink MIMO layers to reduce UE power consumption.	7 BWP Spatial Adaptation (Low-Frequency TDD)
Enhanced function	Uplink SINR compensation for non-coherent rank-1 2T UEs	For 2T UEs working in rank 1, the gNodeB compensates for the PUSCH SINR to increase the MCS index.	8 Uplink SINR Compensation for Non-Coherent Rank-1 2T UEs
Enhanced function	Coordination between UL skipping and BWP switching	When a UE switches between BWP 1 and BWP 2, the UE informs the base station of the BWP switching. This ensures BWP consistency between the base station and the UE and prevents service drops.	9 Coordination Between UL Skipping and BWP Switching
Enhanced function	Joint service experience assurance	When a UE performs a specific service, it exchanges device-pipe information with the base station to jointly guarantee service experience.	10 Joint Service Experience Assurance

4 Device-Pipe Identification

4.1 Principles

Device-pipe identification enables the eNodeB/gNodeB and UEs capable of device-pipe synergy to identify and negotiate with each other. This function supports capability reporting and the configuration of enhanced functions in device-pipe synergy to provide a basis for the enhanced functions.

SA Networking

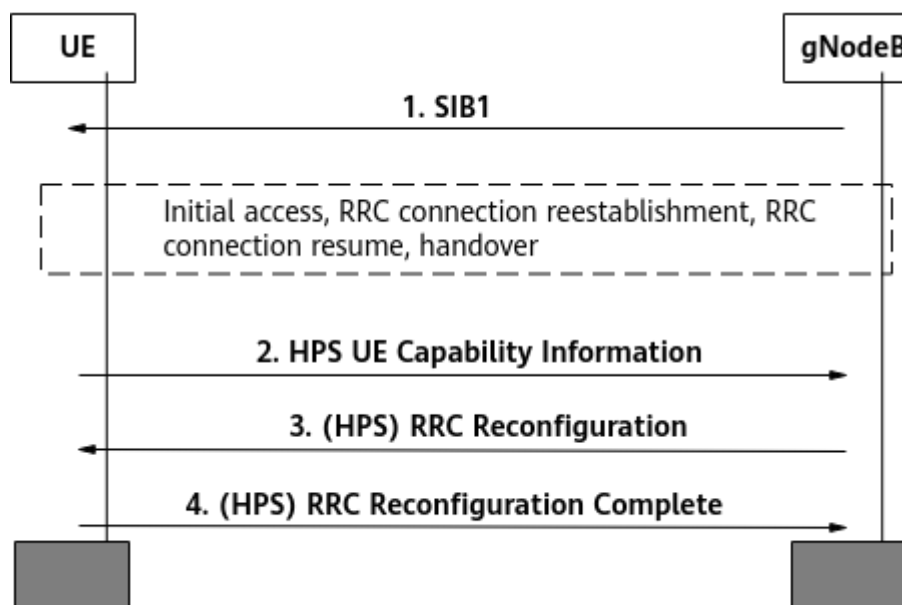
In SA networking, the gNodeB directly negotiates device-pipe capabilities with UEs, and the device-pipe identification switch is configured on the gNodeB. To accommodate device-pipe synergy in NR, Huawei has introduced Huawei Private Signaling (HPS) on the Uu interface. This signaling is exchanged between the gNodeB and UEs for related negotiation and configuration.

The device-pipe identification function is specified by the **SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. This function takes effect only for UEs that support device-pipe identification. The overall process is as follows:

1. The gNodeB broadcasts SIB1 that contains the SIB1 extension field indicating support for device-pipe synergy. Based on this field, a UE determines whether the gNodeB is a Huawei base station that supports device-pipe synergy.
2. The UE reports the support for enhanced functions in device-pipe synergy to the gNodeB through an HPS UE Capability Information message.
3. The gNodeB sends an HPS RRC Reconfiguration message to the UE, notifying the UE of the enabling status and related configuration information of enhanced functions in device-pipe synergy. If the switch for any enhanced function is on, the UE starts to validate the function after receiving the message.
4. After successfully validating enhanced functions in device-pipe synergy, the UE sends an HPS RRC Reconfiguration Complete message to the gNodeB.

Figure 4-1 shows the device-pipe identification process, which can be triggered in the event of initial network access, RRC connection reestablishment, RRC connection resume, or handover.

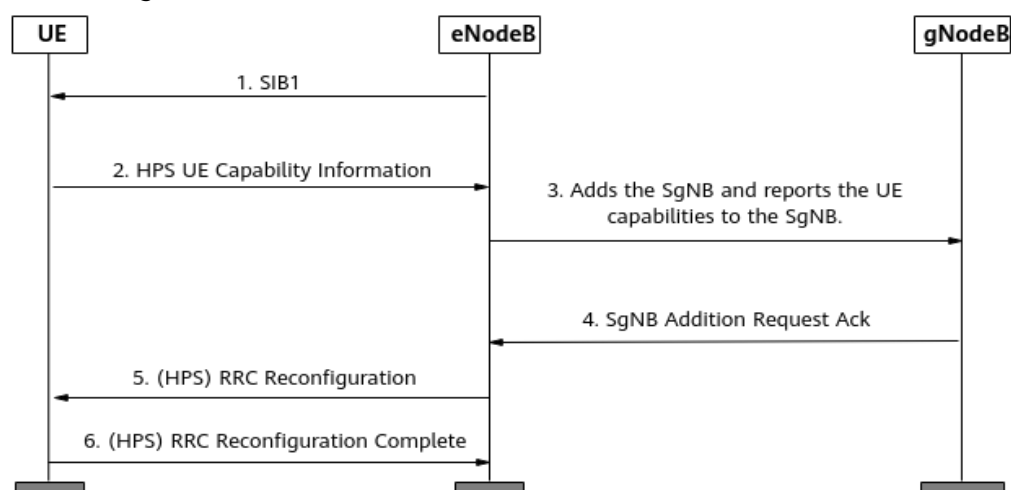
Figure 4-1 Device-pipe identification in SA networking



NSA Networking

In NSA networking, the eNodeB controls the device-pipe identification function, which is specified by the **SpecUeIdentifySwitch** option of the **UeCooperationPara.SpecUserCooperationSwitch** parameter on the eNodeB side. The gNodeB must negotiate device-pipe capabilities with UEs through the eNodeB. [Figure 4-2](#) shows the device-pipe identification process after initial network access in NSA networking.

Figure 4-2 Device-pipe identification after initial network access in NSA networking



1. The eNodeB broadcasts SIB1 that contains the SIB1 extension field indicating support for device-pipe synergy. Based on this field, a UE determines whether the eNodeB is a Huawei base station that supports device-pipe synergy.
2. The UE reports the support for enhanced functions in device-pipe synergy to the eNodeB through an HPS UE Capability Information message.

3. The eNodeB transparently transmits UE capability information to the gNodeB over the X2 interface. The gNodeB then selects a function to be enabled based on the switch status of and UE capability support for enhanced functions in device-pipe synergy.
4. The gNodeB sends an SgNB Addition Request Ack over the X2 interface to notify the eNodeB of the enhanced function switch status.
5. The eNodeB sends an RRC Reconfiguration message (either a standard message or an HPS message) to the UE, instructing the UE to enable enhancement functions. After receiving the signaling, the UE starts to validate the corresponding enhanced functions in device-pipe synergy.
6. After successfully validating enhanced functions in device-pipe synergy, the UE sends an RRC Reconfiguration Complete message to the eNodeB.

4.2 Network Analysis

4.2.1 Benefits

Enhanced functions in device-pipe synergy depend on this function.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

The downlink RB usage increases slightly, the downlink peak traffic volume decreases slightly, and the cell's service drop rate may increase. This is because the gNodeB indicates support for device-pipe synergy by using the SIB1 extension field, and uses HPS for interaction with UEs.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Reference	Description
TD D	Super uplink with independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Device-pipe synergy takes effect only in NUL cells, but not in SUL cells.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	SRS resource set adjustment	SRS_RES_SET_ ADJ_SW option of the NRDUCellSrs. SrsAlgoExtSw itch parameter	<i>Channel Management</i>	When device-pipe identification is enabled, the gNodeB can identify UE types on the network. In this case, after SRS resource set adjustment takes effect, the impacts on the antenna selection capability of antenna-selection UEs for which SRS resource sets have been adjusted are as follows: • For Huawei UEs supporting antenna selection, there is no impact. • For non-Huawei UEs supporting antenna selection, antenna selection is no longer supported. When device-pipe identification is disabled, the gNodeB cannot identify UE types on the network. In this case, after SRS resource set adjustment takes effect, antenna-selection UEs for which SRS resource sets have been adjusted no longer support antenna selection.

4.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Device-pipe synergy requires support from UEs.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

SA Networking

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	SPEC_UE_IDENTITY_SW	To enable device-pipe identification, select this option.

NSA Networking

Table 4-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-2 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	Specified User Cooperation Switch	UeCooperationPara.SpecUserCooperationSwitch	SpecUeIdentifySwitch	To enable device-pipe identification in NSA networking, select this option on the eNodeB side.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- SA networking
//Enabling device-pipe identification
MOD NRDUCELLUECOOP: NrDuCellId=101, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1;
- NSA networking
//Enabling device-pipe identification
MOD UECOOPERATIONPARA: LocalCellId=0, SpecUserCooperationSwitch=SpecUeIdentifySwitch-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- SA networking
//Disabling device-pipe identification
MOD NRDUCELLUECOOP: NrDuCellId=101, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-0;
- NSA networking
//Disabling device-pipe identification
MOD UECOOPERATIONPARA: LocalCellId=0, SpecUserCooperationSwitch=SpecUeIdentifySwitch-0;

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Create a Uu interface tracing task on the MAE-Access. Enable a UE capable of device-pipe synergy to initially access the network. Check whether the RRC_HPS_UE_CAP_INFO message is present in the Uu interface tracing result. If yes, device-pipe identification has taken effect. If no, device-pipe identification has not taken effect.

4.4.3 Network Monitoring

None

5 Compensation for SAR Decrease (Low-Frequency TDD)

5.1 Principles

Specific absorption rate (SAR) indicates the electromagnetic power absorbed or consumed per unit mass by a human body and is measured in units of watts per kilogram (W/kg). SAR reflects the impact of UE-generated electromagnetic energy on human bodies, with a larger SAR value indicating greater impact. Given the same transmit power, a smaller distance between UEs and human skin leads to a larger SAR. To ensure human body safety, the energy generated by a UE transmitting signals must meet corresponding SAR requirements, whereby the UE under the same conditions must reduce the transmit power by a larger amount when in closer proximity human skin. This causes the SRS quality measured by the gNodeB to deteriorate and affects uplink and downlink scheduling. To address the impact of SAR decrease, the compensation for SAR decrease function is introduced.

Compensation for SAR decrease is controlled by the **REDUCE_SAR_COMPENSATION_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. After this function is enabled, the gNodeB notifies UEs that compensation for SAR decrease takes effect. When a UE reduces its transmit power to meet the SAR requirements, the SRS transmit power is raised to increase the SRS quality and improve the uplink and downlink performance. In addition, when the SAR decrease information changes, UEs use MAC messages to report related information. Based on this information, the gNodeB compensates for the channel measurement result to correctly demodulate uplink data.

5.2 Network Analysis

5.2.1 Benefits

Compensation for SAR decrease improves the average downlink throughput of UEs at the cell edge by 3–10%, with the gain closely related to the amount of SAR

decrease. Higher gains are delivered for UEs performing services in closer proximity to the human skin.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

UEs send MAC messages to notify the gNodeB of SAR decrease information whenever the information changes, increasing the number of reporting times. In addition, SRS measurement results need to be compensated during uplink demodulation, causing the uplink RB usage to slightly increase and the peak traffic volume to slightly decrease.

5.3 Requirements

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Enhanced functions in device-pipe synergy can only be enabled after device-pipe identification is enabled.

5.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-FR inter-band CA does not work with compensation for SAR decrease.
Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-band CA does not work with compensation for SAR decrease.

5.3.2.3 Function Impacts

None

5.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Device-pipe synergy requires support from UEs.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Specified UE Cooperation Switch	NRDUCellUeCoop <i>.SpecUeCooperati</i> <i>onSwitch</i>	REDUCE_SAR_CO MPENSATION_SW	To enable compensation for SAR decrease, select this option.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling compensation for SAR decrease  
MOD NRDUCELLUECOOP: NrDuCellId=101,  
SpecUeCooperationSwitch=REDUCE_SAR_COMPENSATION_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling compensation for SAR decrease  
MOD NRDUCELLUECOOP: NrDuCellId=101,  
SpecUeCooperationSwitch=REDUCE_SAR_COMPENSATION_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

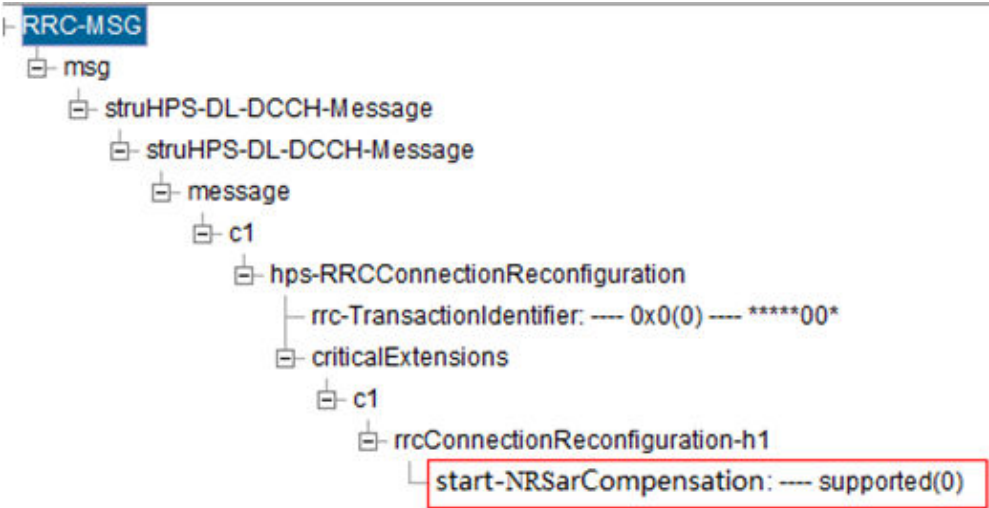
Create a Uu interface tracing task on the MAE-Access. Enable a UE capable of device-pipe synergy to initially access the network. Check that RRC_HPS_CONN_RECFG is displayed in the Uu interface tracing result, as shown in [Figure 5-1](#).

Figure 5-1 RRC_HPS_CONN_RECFG

26	2020-04-18 09:52:39 (508)	RRC_HPS_UE_CAP_INFO
27	2020-04-18 09:52:39 (517)	RRC_RECFG_CMP
28	2020-04-18 09:52:39 (519)	RRC_HPS_CONN_RECFG
29	2020-04-18 09:52:39 (525)	RRC_DL_INFO_TRANSF

If the value of the **start-NRSarCompensation** field (shown in [Figure 5-2](#)) is **1** in the RRC_HPS_CONN_RECFG message, compensation for SAR decrease has taken effect. Otherwise, compensation for SAR decrease has not taken effect.

Figure 5-2 start-NRSarCompensation field



5.4.3 Network Monitoring

This function may increase the downlink throughput for handheld terminals at the cell edge. On the gNodeB side, the average downlink UE throughput can be observed using **User Downlink Average Throughput (DU)**.

6 Fast RRC Connection Release

6.1 Principles

This section mainly describes the fast RRC connection release function for UEs with the **releasePreference-r15** capability. For details about fast RRC connection release for UEs that support the **releasePreference-r16** capability defined in 3GPP Release 16, see *UE Power Saving*.

Fast RRC connection release is controlled by the **RRC_RELEASE_QUICK_SW** option of the **NRCellAlgoSwitch.UePwrSavingSwitch** parameter. When this function takes effect and a UE detects that its RRC connection can be released, the UE proactively sends a UEAssistanceInformation message to release the RRC connection and switch to RRC_IDLE or RRC_INACTIVE, instead of waiting for the expiry of the inactivity timer on the gNodeB side. After receiving the message, the gNodeB releases the UE's RRC connection, reducing UE power consumption. For details about the RRC_INACTIVE state, see *SA Mobility Management in Inactive Mode*.

After receiving a UEAssistanceInformation message from a UE that requests an RRC connection release for switching to the RRC_INACTIVE state, the gNodeB releases the UE to the RRC_IDLE state instead of RRC_INACTIVE when any of the following conditions is met for the cell, UE, or core network:

- The **RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.InactiveStrategySwitch** parameter is deselected for the cell.
- The cell is not configured with a RAN area code (specified by **NRCell.RanNotificationAreald**).
- The gNodeB load is high.
- The core network does not deliver the *NgcoreNwAssistInfo* configuration to the gNodeB.
- The core network delivers the *NgcoreNwAssistInfo* configuration that carries the *MicoModelnd* information to the gNodeB.
- The UE does not support the RRC_INACTIVE state.
- No DRB is set up for the UE.

- The UE has a voice bearer.
- The UE is a Cloud VR UE.

6.2 Network Analysis

6.2.1 Benefits

This function can reduce the power consumption of the UE communication module by up to 6.5%. The gains must be observed and evaluated on the UE side.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

This function may cause the following: The number of times UEs in a cell enter the RRC_IDLE or RRC_INACTIVE state increases, the number of times RRC connections resume increases, the number of UEs in the RRC_CONNECTED state decreases, the RRC connection setup success rate slightly decreases, and the number of attempted or successful UE context setups increases. In addition, when UEs running services proactively initiate fast RRC connection release, the service delay may increase. For details about the impact of the RRC_INACTIVE state on the network, see *SA Mobility Management in Inactive Mode*. The specific impact is closely related to the RRC connection release policy of the UEs. If UEs frequently report messages for fast RRC connection release, the CPU load of the main control board increases.

- The number of RRC connection setup requests increases with the number of times UEs enter the RRC_IDLE state. The related counters are **N.RRC.SetupReq.Att**, **N.RRC.Setup**, and **N.RRC.SetupReq.Succ**.
- The counters related to the number of RRC connection resume times are **N.RRC.ResumeReq.Att**, **N.RRC.Resume**, and **N.RRC.ResumeReq.Succ**.
- The counters related to the number of UEs in the RRC_CONNECTED state are **N.User.RRCConn.Max**, **N.User.RRCConn.Avg**, and **N.User.NsaDc.PSCell.Avg**.
- The RRC connection setup success rate is measured by **RRC Setup Success Rate (CU, Inactive)**.
- The number of UE context setup attempts in a cell is measured by **N.UECntx.Est.Att**.
- The number of successful UE context setups in a cell is measured by **N.UECntx.Est.Succ**.

6.3 Requirements

6.3.1 Licenses

There are no license requirements for basic functions.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR TDD	Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Enhanced functions in device-pipe synergy can only be enabled after device-pipe identification is enabled.

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
SA mobility management in inactive mode	RRC_INACTIVE_SWITCH option of the NRCCellAlgoSwitch.InactiveStrategySwitch parameter	<i>SA Mobility Management in Inactive Mode</i>	When the RRC_INACTIVE state function is enabled, fast RRC connection release allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state.

6.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Device-pipe synergy requires support from UEs.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

[Table 6-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-1 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	UE Power Saving Switch	NRCellAlgoSwitch.UePwrSavingSwitch	RRC_RELEASE_Q UICK_SW	To enable fast RRC connection release, select this option.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling fast RRC connection release  
MOD NRCELLALGOSWITCH: NrCellId=101, UePwrSavingSwitch=RRC_RELEASE_QUICK_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling fast RRC connection release  
MOD NRCELLALGOSWITCH: NrCellId=101, UePwrSavingSwitch=RRC_RELEASE_QUICK_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

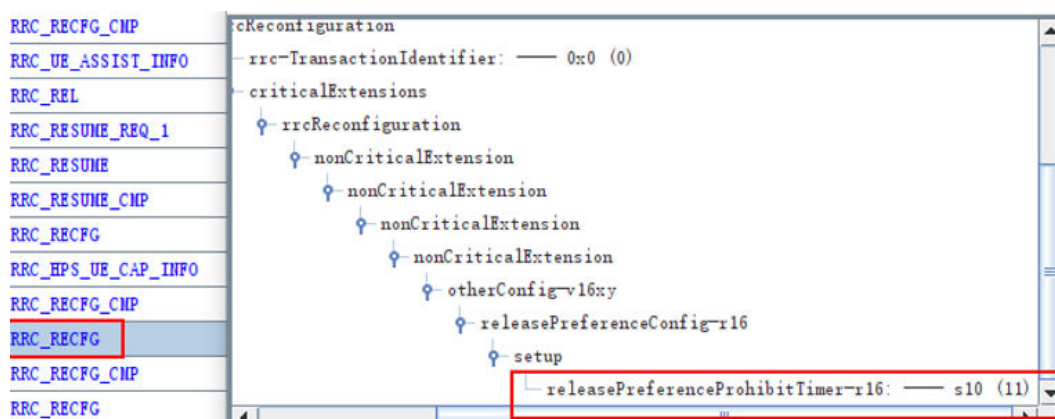
NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Create a Uu interface tracing task on the MAE-Access. Enable a UE capable of device-pipe synergy to initially access the network. Check the Uu interface tracing result. If the RRC_RECFCG message carries the **releasePreferenceConfig** field in which **releasePreferenceProhibitTimer-r16** is set to a value other than 0, as shown in [Figure 6-1](#), fast RRC connection release has taken effect. In other cases, fast RRC connection release has not taken effect.

Figure 6-1 releasePreferenceConfig field



6.4.3 Network Monitoring

Subscribe to CHR logs. If the value of the **RelTrigger** field in the **PRIVATE_UE_RELEASE** event block is **UE**, the UE proactively requests an RRC connection release for switching to the RRC_IDLE state. After the fast RRC

connection release function takes effect, the number of times that this event block is triggered increases.

Subscribe to CHR logs. If the value of the **RelTrigger** field in the **PRIVATE_RRC_RELEASE_SUSPEND** event block is **UE**, the UE proactively requests an RRC connection release for switching to the RRC_INACTIVE state. After the fast RRC connection release function takes effect, the number of times that this event block is triggered increases.

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

7 BWP Spatial Adaptation (Low-Frequency TDD)

7.1 Principles

This section describes the BWP spatial adaptation function for UEs with the **maxLayersMIMO-Adaptation-r15** capability. For details about BWP spatial adaptation for UEs that support the **maxLayersMIMO-Adaptation-r16** capability, see *UE Power Saving*.

The BWP spatial adaptation function is controlled by the **MIMO_BWP_ADAPT_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. When this function takes effect, a UE requiring power saving reports the maximum number of downlink MIMO layers to be used to the gNodeB. The gNodeB selects an appropriate power saving BWP for the UE. The UE shuts down some receive antennas after switching to this BWP, and the gNodeB performs downlink scheduling based on the UE-reported maximum number of downlink MIMO layers to reduce UE power consumption.

7.2 Network Analysis

7.2.1 Benefits

This function can reduce the power consumption of the UE communication module by up to 5.5%. The gains must be observed and evaluated on the UE side.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

On the UE side, this function may decrease the throughput of UEs supporting BWP2 and increase the service delay of such UEs.

On the gNodeB side, this function may have the following negative impacts:

- The average cell throughput and average UE throughput decrease.
- The delay increases.
- The HARQ retransmission rate and BLER increase.
- The rank proportion increases or decreases.

In addition, when BWP spatial adaptation takes effect, UEs with power saving BWP in effect automatically shut down some receive antennas, exacerbating the negative impact.

- The average cell throughput is measured using **Cell Downlink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**.
- The average UE throughput is measured using **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**.
- Delay

First Packet Delay in RLC Empty Buffer = $\frac{N.Traffic.DL.RlcFirstPktDelay.Time}{N.QoS.DL.RlcFirstPktDelay.Num}$

Packet Delay in MAC = $\frac{N.QoS.DL.PktDelayAirInterface.Time}{N.QoS.DL.PktDelayAirInterface.Num}$

Packet Delay in RLC = $\frac{N.QoS.DL.PktDelaygNBDU.Time}{N.QoS.DL.PktDelaygNBDU.Num}$

Packet Delay in DU = $\frac{N.QoS.DL.PktDelayAirInterface.Time}{N.QoS.DL.PktDelayAirInterface.Num} + \frac{N.QoS.DL.PktDelaygNBDU.Time}{N.QoS.DL.PktDelaygNBDU.Num}$
- Uplink HARQ retransmission rate

Uplink HARQ retransmission rate = $\frac{(N.UL.SCH.HalfPiBPSK.TB.Retrans + N.UL.SCH.QPSK.TB.Retrans + N.UL.SCH.16QAM.TB.Retrans + N.UL.SCH.64QAM.TB.Retrans + N.UL.SCH.256QAM.TB.Retrans)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB + N.UL.SCH.HalfPiBPSK.TB.Retrans + N.UL.SCH.QPSK.TB.Retrans + N.UL.SCH.16QAM.TB.Retrans + N.UL.SCH.64QAM.TB.Retrans + N.UL.SCH.256QAM.TB.Retrans)} \times 100\%$
- Downlink HARQ retransmission rate

Downlink HARQ retransmission rate = $\frac{(N.DL.SCH.QPSK.TB.Retrans + N.DL.SCH.16QAM.TB.Retrans + N.DL.SCH.64QAM.TB.Retrans + N.DL.SCH.256QAM.TB.Retrans)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB + N.DL.SCH.QPSK.TB.Retrans + N.DL.SCH.16QAM.TB.Retrans + N.DL.SCH.64QAM.TB.Retrans + N.DL.SCH.256QAM.TB.Retrans)} \times 100\%$
- BLER

MAC Uplink BLER = $\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$

MAC Downlink BLER = $\frac{(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$

$$\frac{\text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB}}{100\%}$$

- Rank proportion

$$\text{Rank 1 proportion of downlink TBs} = \frac{\text{N.PDSCH.NbrTbDL.Rank1}}{(\text{N.PDSCH.NbrTbDL.Rank1} + \dots + \text{N.PDSCH.NbrTbDL.Rank8})} \times 100\%$$

$$\text{Rank 2 proportion of downlink TBs} = \frac{\text{N.PDSCH.NbrTbDL.Rank2}}{(\text{N.PDSCH.NbrTbDL.Rank1} + \dots + \text{N.PDSCH.NbrTbDL.Rank8})} \times 100\%$$

$$\text{Rank 3 proportion of downlink TBs} = \frac{\text{N.PDSCH.NbrTbDL.Rank3}}{(\text{N.PDSCH.NbrTbDL.Rank1} + \dots + \text{N.PDSCH.NbrTbDL.Rank8})} \times 100\%$$

$$\text{Rank 4 proportion of downlink TBs} = \frac{\text{N.PDSCH.NbrTbDL.Rank4}}{(\text{N.PDSCH.NbrTbDL.Rank1} + \dots + \text{N.PDSCH.NbrTbDL.Rank8})} \times 100\%$$

7.3 Requirements

7.3.1 Licenses

There are no license requirements for basic functions.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR TDD	Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Enhanced functions in device-pipe synergy can only be enabled after device-pipe identification is enabled.
NR TDD	Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePowerSaving.BwpPowerSavingSw parameter	<i>UE Power Saving</i>	BWP spatial adaptation can only be enabled after the power saving BWP function is enabled.

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Device-pipe synergy requires support from UEs.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

[Table 7-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-1 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	MIMO_BWP_ADAPT_SW	To enable BWP spatial adaptation, select this option.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling BWP spatial adaptation  
MOD NRDUCELLUECOOP: NrdCellId=101, SpecUeCooperationSwitch=MIMO_BWP_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling BWP spatial adaptation  
MOD NRDUCELLUECOOP: NrdCellId=101, SpecUeCooperationSwitch=MIMO_BWP_ADAPT_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Create a Uu interface tracing task on the MAE-Access. Enable a UE capable of device-pipe synergy to initially access the network. Check the Uu interface tracing result. If the **hps-mimoDLBwpId** field in the RRC_HPS_CONN_RECFG message is set to **2**, BWP spatial adaptation has taken effect. Otherwise, this function has not taken effect.

7.4.3 Network Monitoring

After BWP spatial adaptation is enabled, the average cell throughput, average UE throughput, delay, HARQ retransmission rate, BLER, and rank proportion may change on the network. [7.2.2 Impacts](#) provides related counters.

The following counters can also be used for monitoring BWP spatial adaptation.

Counter ID	Counter Name	Monitoring Purpose
1911825117	N.Bwp2.Dur.Total	This counter measures the total duration of UEs using BWP2 in a cell.
1911825121	N.BWP.ThpVol.DL	These counters can be used to measure the average throughputs of UEs working in BWP0 and BWP1 in a cell. UEs working in BWP0 are those working in the initial BWP.
1911825125	N.BWP.ThpVol.DL.LastSlot	
1911825128	N.BWP.ThpTime.DL.RmvLastSlot	
1911825129	N.BWP.ThpVol.UL	
1911825123	N.BWP.ThpVol.UE.UL.SmallPkt	
1911825127	N.BWP.ThpTime.UE.UL.RmvSmallPkt	

Average downlink throughput of UEs working in BWP0 and BWP1 =

$$\frac{(\text{N.BWP.ThpVol.DL} - \text{N.BWP.ThpVol.DL.LastSlot}) \times 1000}{\text{N.BWP.ThpTime.DL.RmvLastSlot}}$$

Average uplink throughput of UEs working in BWP0 and BWP1 =

$$\frac{(\text{N.BWP.ThpVol.UL} - \text{N.BWP.ThpVol.UE.UL.SmallPkt}) \times 1000}{\text{N.BWP.ThpTime.UE.UL.RmvSmallPkt}}$$

NOTE

- As UE power saving gains are closely related to the BWP capability of UEs in a cell as well as the traffic model, they cannot be evaluated using the **N.Bwp2.Dur.Total** counter. Instead, such gains must be observed and evaluated on the UE side.
- The average UE throughput (**User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**) is calculated based on all UEs in a cell, and are different from the average throughputs of UEs working in BWP0 and BWP1.
- The average throughputs of UEs working in BWP0 and BWP1 are related to the traffic model and the measured values may be 0.

8 Uplink SINR Compensation for Non-Coherent Rank-1 2T UEs

8.1 Principles

The transmit power of the PUSCH for a non-coherent 2T UE working in rank 1 is higher than that of the SRS. As a result, the PUSCH MCS index of the UE is relatively low at the cell edge, leading to relatively low UE throughput. To increase the throughput of cell edge UEs, uplink SINR compensation for non-coherent rank-1 2T UEs is introduced.

When device-pipe identification is enabled, if the gNodeB identifies a UE that supports device-pipe synergy (based on the **support-EnhancedCDD-TransmitPower** field) and is a 2T UE (that is, the value of **maxNumberSRS-Ports-PerResource** in the UE-reported capability information is **n2**) using rank 1 in the uplink, the gNodeB compensates for the SINR when selecting the MCS index for the PUSCH. The **NRDUCellUeCoop.UlRank1TwoTxSinrAdjust** parameter controls the amount of compensation. This function increases the MCS index for the PUSCH of UEs using rank 1 and the uplink throughput of cell edge UEs.

8.2 Network Analysis

8.2.1 Benefits

After this function is enabled, the uplink MCS index and average uplink throughput of cell edge UEs, and the average uplink cell throughput increase. The specific gains are related to the proportion of cell edge UEs.

NOTE

- Counters related to uplink MCS index: **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.31**
- Average uplink UE throughput: **User Uplink Average Throughput (DU)**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

The network impacts of this function depend on the value of the **NRDUCellUeCoop.UlRank1TwoTxSinrAdjust** parameter. If this parameter is set to an excessively large value, the uplink BLER increases.

8.3 Requirements

8.3.1 Licenses

There are no license requirements for basic functions.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR TDD	Device-pipe identification	SPEC_UE_IDENTITY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Enhanced functions in device-pipe synergy can only be enabled after device-pipe identification is enabled.

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

None

8.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Device-pipe synergy requires support from UEs.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

RAT	Parameter Name	Parameter ID	Option	Setting Notes
TDD	Uplink Rank1 Two TX SINR Adjustment Value	NRDUCellUeCo op.UlRank1Two TxSinrAdjust	None	Set this parameter to a value other than 0 based on the actual network conditions.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink SINR compensation for non-coherent rank-1 2T UEs  
MOD NRDUCELLUECOOP: NrDuCellId=101, UIRank1TwoTxSinrAdjust=10;
```

Deactivation Command Examples

```
//Disabling uplink SINR compensation for non-coherent rank-1 2T UEs  
MOD NRDUCELLUECOOP: NrDuCellId=101, UIRank1TwoTxSinrAdjust=0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

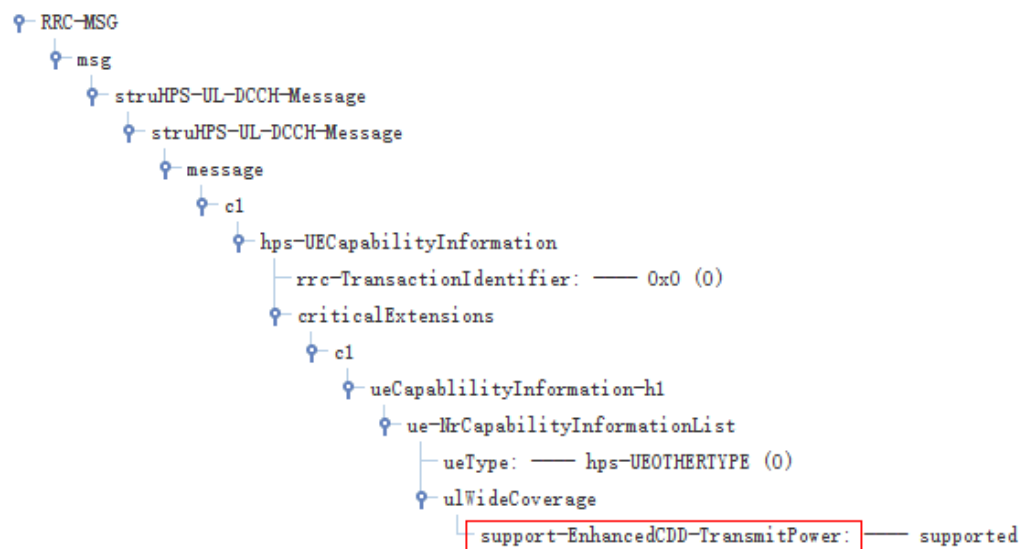
NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Create a Uu interface tracing task on the MAE-Access. Enable a UE capable of device-pipe synergy to initially access the network. Start a Uu interface tracing task. Observe RRC_RECFG signaling. If there is an RRC_HPS_CONN_RECFG message carrying **support-EnhancedCDD-TransmitPower**, as shown in [Figure 8-1](#), this function has taken effect. If there is not, this function has not taken effect.

Figure 8-1 support-EnhancedCDD-TransmitPower field



8.4.3 Network Monitoring

Verify the following to obtain the gains of this function:

- **User Uplink Average Throughput (DU)** increases.
- **Cell Uplink Average Throughput (DU)** increases.

9 Coordination Between UL Skipping and BWP Switching

9.1 Principles

3GPP specifications have proposed UL skipping for UE power saving. With the UL skipping function, for a UE supporting UL skipping, sending padding packets is not required if the UE has no data to transmit after it receives UL grants from the base station. This saves UE power. However, when a UE with UL skipping in effect switches between BWP 1 and BWP 2, if the UE has no data to transmit after BWP switching, it cannot send uplink padding packets to inform the base station of the new BWP. This may cause BWP difference between the UE and the base station, which leads to UE service drops and an increase in the service drop rate of the cell. As such, coordination between UL skipping and BWP switching is introduced. This function is controlled by the **ULSKIP_BWP_COORD_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter.

When the device-pipe identification function is enabled, the gNodeB identifies UEs supporting device-pipe synergy based on the **hps-support-dciSwitchBwpForULSkip** field. After UL skipping takes effect for a UE with the capability of this field, coordination between UL skipping and BWP switching can take effect to ensure BWP consistency between the base station and the UE after BWP switching and prevent UE service drops.

9.2 Network Analysis

9.2.1 Benefits

After this function is enabled, the service drop rate in the cell may decrease.

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

There is no impact on the network.

9.3 Requirements

9.3.1 Licenses

None

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Coordination between UL skipping and BWP switching can be enabled only when device-pipe identification is enabled.
UL skipping	UPLINK_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	Coordination between UL skipping and BWP switching can be enabled only when UL skipping is enabled.
Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Coordination between UL skipping and BWP switching can be enabled only when power saving BWP is enabled.

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

None

9.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Others

This function requires that UEs support both UL skipping and coordination between UL skipping and BWP switching. For coordination between UL skipping and BWP switching, UEs must have the **hps-support-dciSwitchBwpForULSkip** capability. For UL skipping, UEs must have the **SkipUplinkTxDynamic**, **enhancedSkipUplinkTxDynamic-r16**, or **enhancedSkipUplinkTxDynamic-v1660** capability.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 9-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	To enable coordination between UL skipping and BWP switching for a cell, select the ULSKIP_BWP_COORD_SW option.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling coordination between UL skipping and BWP switching  
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=ULSKIP_BWP_COORD_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling coordination between UL skipping and BWP switching  
MOD NRDUCELLUECOOP: NrDuCellId=0, SpecUeCooperationSwitch=ULSKIP_BWP_COORD_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

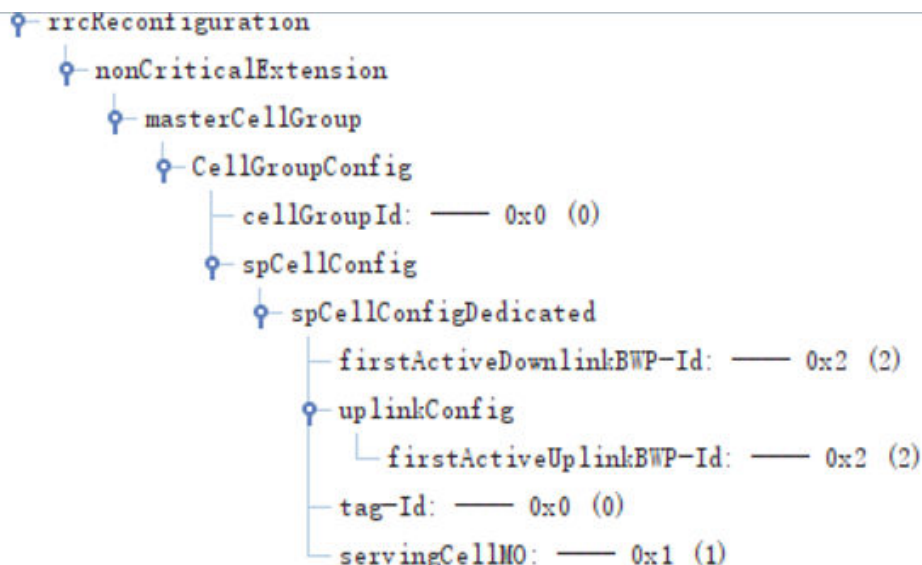
9.4.2 Activation Verification

This function has taken effect if both of the following are true:

- Signaling observation

- a. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.
- b. In the navigation tree on the left, choose **Uu Interface Trace**.
- c. On the displayed **Uu Interface Trace** page, check the RRC_RECFC message. The *rrcReconfiguration* IE does not carry **firstActiveUplinkBWP-Id** or **firstActiveDownlinkBWP-Id** with the value of 2.

Figure 9-1 firstActiveUplinkBWP-Id and firstActiveDownlinkBWP-Id fields



- Counter observation
The value of the **N.Bwp2.Dur.Total** counter is not 0.

9.4.3 Network Monitoring

After this function is enabled, monitor the running status of this function by observing value changes in the service drop rate of the cell.

10 Joint Service Experience Assurance

10.1 Principles

The network side cannot accurately perceive the UE experience across different services, while the UE cannot perceive the network status. As a result, the base station and UE cannot jointly ensure service experience. To address this, joint service experience assurance is introduced. This function includes the following subfunctions:

- **Service Experience Awareness**
- **Service Identification-based Assurance**
- **Service Experience Collaboration**
- **Cell Usage Scenario Indication by the Base Station**

This function requires that the base station and UEs complete the device-pipe identification procedure. For details about this procedure, see [4 Device-Pipe Identification](#). This section describes the processing exclusive to joint service experience assurance in comparison to basic device-pipe identification.

1. UE capability reporting: The UE sends an HPS UE Capability Information message to the gNodeB to report whether the UE itself supports QoS reporting or cell attribute parsing.
 - QoS reporting: The UE indicates this capability through the **supportServiceQosType** field in *HPS-HighPerformance*.
 - Cell attribute parsing: The UE indicates this capability through the **supportCellModeNotify** field in *HPS-CellModeNotify-Capability*.
2. Function enabling status notification: The gNodeB sends an HPS RRC Reconfiguration message to notify the UE of the enabling status of related functions and configuration information. If the related functions are enabled, they take effect after the UE that supports these functions receives this message.
3. Function activation response: If service experience awareness or service experience collaboration successfully takes effect on the UE, the UE sends an HPS RRC Reconfiguration Complete message as a response to the gNodeB.

Service Experience Awareness

UEs capable of QoS reporting send the categories of ongoing services (such as real-time low-traffic, real-time medium-traffic, or non-real-time medium-traffic) to the base station. This enhances the capability of the network side to be aware of the service experience of the UEs. This function is controlled by the **SPEC_UE_EXP_AWARE_SW** option of the **NRCCellAlgoSwitch.SpecUeCollabSwitch** parameter.

Service Identification-based Assurance

After service experience awareness is enabled, service identification-based assurance enables a base station to identify the service types and provides differentiated service assurance after obtaining the service types from UEs.

Service identification-based assurance can be enabled when all the following conditions are met:

- **gNBSchAdaptToTfcMdl.SchAdaptToTfcModelPol** is set to **TAI**.
- **gNBServExpGroup.Nr5qi** is mapped to **gNBServExpGroup.EquivalentQci**.
- QCI-specific assurance functions are enabled based on the value of **gNBServExpGroup.EquivalentQci**. For details about QCI-specific assurance functions, see *Service-differentiated Experience Guarantee*.

If a bearer is configured with a QCI for differentiated QoS management for RedCap UEs (specified by **gNBOP5qiConfig.RedCapQci**) and requires service assurance using the equivalent QCI for service identification-based assurance (specified by **gNBServExpGroup.EquivalentQci**), QCI mapping policy optimization must be enabled by selecting the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo.QciAlgoOptSw** parameter.

- If **gNBOP5qiConfig.RedCapQci** is set to **65535** and **gNBOP5qiConfig.Qci** is set to a value other than **65535**, the equivalent QCI for service identification-based assurance can be used for the bearers of RedCap UEs.
- If **gNBOP5qiConfig.RedCapQci** is set to a value other than **65535** and **gNBOP5qiConfig.Qci** is set to **65535**, the equivalent QCI for service identification-based assurance can be used for the bearers of non-RedCap UEs.

Service Experience Collaboration

When service experience collaboration is enabled, UEs capable of QoS reporting request information from the base station in specific scenarios (such as scenarios with poor uplink service experience due to high interference). The base station will send the UEs an **HPS_DL_TRANSF_MSG** message containing load- and interference-based service recommendation indexes for the serving cell and neighboring cells, so the UEs can perform service assurance based on these indexes. This function is controlled by the **SPEC_UE_EXP_COLLAB_SW** option of the **NRCCellAlgoSwitch.SpecUeCollabSwitch** parameter.

Cell Usage Scenario Indication by the Base Station

A base station can configure different cell attributes based on actual usage scenarios of cells, such as subway, airport, or scenic spot, and indicate cell

attributes to UEs that are capable of parsing cell attributes through the **hps-CellMode** field in *HPS-UserExperienceConfig*. UEs perform differentiated assurance on services based on the cell attribute indication. For example, UEs select proper cells to camp on, optimize service experience of an app, optimize the network search near a scenic spot or a subway, and buffer more videos before boarding a plane or subway.

Cell attributes can be configured using the **NRCellServExp.NrCellUsageScenario** parameter. If this parameter is set to **NotConfig**, this function does not take effect. If this parameter is set to other values, the base station indicates the corresponding cell attribute information to UEs that are capable of parsing cell attributes.

10.2 Network Analysis

10.2.1 Benefits

After this function is enabled, service suspension of specific UEs can be reduced, improving UE service experience.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

The impacts on the network are as follows:

- When the service experience awareness function takes effect, the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) of the cell may increase.
- When the service experience collaboration function takes effect, the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) and downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) of the cell may increase.
- The network impacts of service identification-based assurance are related to the effective QCI-specific assurance functions. For details, see the corresponding "Impacts" sections of related assurance functions in *Service-differentiated Experience Guarantee*.
- Cell usage scenario indication by the base station: None

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	- Service experience awareness and service experience collaboration can be enabled only when device-pipe identification is enabled. - Cell usage scenario indication by the base station can be enabled only when device-pipe identification is enabled.
Service experience awareness	SPEC_UE_EXP_AWARE_SW option of the NRCCellAlgoSwitch.SpecUeCollabSwitch parameter	<i>Device-Pipe Synergy</i>	Service identification-based assurance can be enabled only when service experience awareness is enabled.
Service Type	gNBQciBearer.ServiceType	None	If gNBServExpGroup.ServiceType is set to EMBB_LOW_TRAFFIC , service identification-based assurance can be enabled only when gNBQciBearer.ServiceType is set to NORMAL .

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Differentiated QoS management for RedCap UEs	gNBOp5qiConfig.RedCapQci set to a value other than 65535	<i>RedCap</i>	• If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgorithmOptSw parameter is deselected and the corresponding gNBOp5qiConfig MO is configured for a bearer, the equivalent QCI for service identification-based assurance (specified by gNBServExpGroup.EquivalentQci) of the bearer and service identification-based assurance do not take effect. • If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgorithmOptSw parameter is selected and a QCI for differentiated QoS management for RedCap UEs (specified by gNBOp5qiConfig.RedCapQci) has been configured for a bearer: – If gNBOp5qiConfig.RedCapQci is set to 65535, service identification-

RAT	Function Name	Function Switch	Reference	Description
				based assurance does not take effect for non-RedCap UEs. - If gNBOP5qiConfig.RedCapQci is set to a value other than 65535: - If gNBOP5qiConfig.Qci is set to 65535, service identification-based assurance does not take effect for RedCap UEs. - If gNBOP5qiConfig.Qci is set to a value other than 65535, service identification-based assurance does not take effect for all UEs.

10.3.3 Hardware

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

10.3.4 Others

- Service experience awareness, service identification-based assurance, and service experience collaboration: UEs must support QoS reporting. In other words, UEs have the **support-ServiceQosType** capability.
- Cell usage scenario indication by the base station: UEs must be capable of parsing cell attributes. In other words, UEs have the **supportCellModeNotify** capability.

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Table 10-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment. For details about data preparation of functions related to differentiated service assurance, see the corresponding "Data Preparation" sections in *Service-differentiated Experience Guarantee*.

Table 10-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Specified UE Collaboration Switch	NRCellAlgoSwitch.SpecUeCollabSwitch	Select the SPEC_UE_EXP_AWARE_SW option if the service experience awareness function is required in a cell. Select the SPEC_UE_EXP_COLLAB_SW option if the service experience collaboration function is required in a cell.

Parameter Name	Parameter ID	Setting Notes
Schedule Adapt to Traffic Model Policy	gNBSchAdaptToTfcMdl.SchAdaptToTfcModelPol	Set this parameter to TAI if service identification-based assurance is required in a cell.
Qci Algorithm Optimization Switch	gNodeBAlgo.QciAlgoOptSw	Select the QCI_MAP_POLICY_OPT_SW option if a bearer is configured with a QCI specified by either gNBOp5qiConfig.Qci or gNBOp5qiConfig.RedCapQci set to 65535 and requires service assurance using the equivalent QCI for service identification-based assurance (specified by gNBServExpGroup.EquivalentQci).
NR Cell Usage Scenario	NRCellServExp.NrCellUsageScenario	Set this parameter based on the actual usage scenario.
Service Type Group ID	gNBServExpGroup.ServiceTypeGroupId	Set this parameter based on the network plan.
NR User Networking Option	gNBServExpGroup.NrUserNetworkingOption	Set this parameter based on the network plan.
NR 5G QoS Indicator	gNBServExpGroup.Nr5qi	Set this parameter based on the network plan.
Service Type	gNBServExpGroup.ServiceType	Set this parameter based on the network plan.
Equivalent QoS Class Identifier	gNBServExpGroup.EquivalentQci	Set this parameter based on the network plan.

10.4.1.2 Using MML Commands

Before using MML commands, refer to [10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

For details about MML-based configurations of functions related to differentiated service assurance, see the corresponding "Using MML Commands" sections in *Service-differentiated Experience Guarantee*.

Activation Command Examples

```
//Enabling the service experience awareness function
MOD NRCELLALGOSWITCH: NrCellId=0, SpecUeCollabSwitch=SPEC_UE_EXP_AWARE_SW-1;
//Enabling service identification-based assurance
MOD GNBSCHADAPTTOTFCMDL: SchAdaptToTfcModelPol=TAI;
ADD GNBSERVEXPGROUP: ServiceTypeGroupId=1, NrUserNetworkingOption=SA, Nr5qi=9,
ServiceType=EMBB_LOW_TRAFFIC, EquivalentQci=201;
//(Optional) Enabling QCI mapping policy optimization if a bearer is configured with a QCI specified by
either gNBOP5qiConfig.Qci or gNBOP5qiConfig.RedCapQci set to 65535 and requires service assurance
using the equivalent QCI for service identification-based assurance (specified by
gNBSErvExpGroup.EquivalentQci).
MOD GNODEBALGO: QciAlgoOptSw=QCI_MAP_POLICY_OPT_SW-1;
//Enabling service experience collaboration
MOD NRCELLALGOSWITCH: NrCellId=0, SpecUeCollabSwitch=SPEC_UE_EXP_COLLAB_SW-1;
//Enabling cell usage scenario indication by the base station (using the railway scenario as an example)
MOD NRCELLSERVEXP: NrCellId=0, NrCellUsageScenario=RAILWAY;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling service identification-based assurance
MOD GNBSCHADAPTTOTFCMDL: SchAdaptToTfcModelPol=DEFAULT;
//Disabling QCI mapping policy optimization
MOD GNODEBALGO: QciAlgoOptSw=QCI_MAP_POLICY_OPT_SW-0;
//Disabling the service experience awareness function
MOD NRCELLALGOSWITCH: NrCellId=0, SpecUeCollabSwitch=SPEC_UE_EXP_AWARE_SW-0;
//Disabling the service experience collaboration function
MOD NRCELLALGOSWITCH: NrCellId=0, SpecUeCollabSwitch=SPEC_UE_EXP_COLLAB_SW-0;
//Disabling cell usage scenario indication by the base station
MOD NRCELLSERVEXP: NrCellId=0, NrCellUsageScenario=NOT_CONFIG;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

1. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.
2. In the navigation tree on the left, choose **Uu Interface Trace**.
3. Enable a UE capable of device-pipe synergy to initially access the network.
4. On the **Uu Interface Trace** page, check for the RRC_HPS_UE_CAP_INFO message. If the RRC_HPS_UE_CAP_INFO message is displayed:

- When the value of the **support-ServiceQosType** field is **supported**, the UE supports joint service experience assurance. Otherwise, the UE does not support joint service experience assurance.
 - When the value of the **supportCellModeNotify** field is **supported**, the UE supports parsing cell attributes. Otherwise, the UE does not support parsing cell attributes.
5. If the UE supports joint service experience assurance, check for the RRC_HPS_CONN_RECFG message on the **Uu Interface Trace** page. If the message is displayed:
- When the value of the **hps-QosRequirement** field in the RRC_HPS_CONN_RECFG message is **enabled**, service experience awareness has taken effect.
 - When the value of the **hps-ResourceRequirement** field in the RRC_HPS_CONN_RECFG message is **enabled**, service experience collaboration has taken effect.
 - When the *HPS-UserExperienceConfig* IE in the RRC_HPS_CONN_RECFG message contains the **hps-CellMode** field, cell usage scenario indication by the base station has taken effect.

10.4.3 Network Monitoring

The running status of this function can be monitored based on the changes in the following counters:

- Uplink PRB usage in a cell: **Uplink Resource Block Utilizing Rate (DU)**
- Downlink PRB usage in a cell: **Downlink Resource Block Utilizing Rate (DU)**

11 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

12 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

13 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

14 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Super Uplink*
- *Carrier Aggregation*
- *UE Power Saving*
- *SA Mobility Management in Inactive Mode*
- *Channel Management*
- *Service-differentiated Experience Guarantee*
- *RedCap*